DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026 : WORKSHEET-1

Subject: Science (086) Biology

Chapter: 2 – CONTROL AND COORDINATION

Date: _____ **Name :**_____ **Class / Sec. X**_____

Section A: Multiple Choice Questions (MCQs)

1. The reflex arc involves:

   a. Brain only b. Muscles c. Spinal cord d. Heart

2. Which hormone lowers the blood glucose level?

   a. Insulin b. Adrenaline c. Thyroxine d. Glucagon

3. Involuntary actions are controlled by:

   a. Cerebrum b. Spinal cord c. Medulla d. Cerebellum

4. Which of the following is a plant hormone?

   a .Adrenaline b. Auxin c. Insulin d. Estrogen

5. What type of movement occurs in plants due to external stimuli?

   a. Growth movement b. Locomotion c. Tropic movement d. Photokinesis

6. Which part of the brain is responsible for posture and balance?

   a. Medulla b. Cerebrum c. Pons d. Cerebellum

7. The junction between two neurons is called:

   a. Axon b. Synapse c. Dendrite d. Myelin

8. Which gland is called the master gland?

   a. Thyroid b. Adrenal c. Pituitary d. Pancreas

9. Adrenaline is secreted during:

   a. Sleep b. Calmness c. Stress d. Digestion

10. The shoot system of plants grows toward:

    a. Water b. Gravity c. Light d. Touch

11. Diabetes is caused due to deficiency of:

    a. Insulin b. Thyroxine c. Estrogen d. Testosterone

12. Coordination in animals is maintained by:

    a. Muscular system b. Circulatory system c. Nervous and endocrine systems d. Skeletal system

13. Which part of the brain regulates heartbeat and breathing?

    a. Cerebrum b. Cerebellum c. Medulla oblongata d. Thalamus

14. Which hormone increases heart rate during emergencies?

a. Adrenaline b. Oxytocin c. Insulin d. Growth hormone

15. Tropic movements in plants are:

a. Growth-based and directional Movement b. Circular c. Non-directional d. growth independent

16. Cytokinins promote:

a. Cell division b. Leaf drop c. Fruit ripening d. Cell elongation

17. The spinal cord is protected by:

a. Skull b. Ribs c. Vertebral column d. Sternum

18. A hormone regulating metabolism is:

a. Adrenaline b. Thyroxine c. Estrogen d. Insulin

19. Which part of neuron receives signals?

a. Axon b. Dendrite c. Synapse d. myelin

20. Geotropism refers to movement in response to:

a. Touch b. Gravity c. Water d. Heat

21. Goitre is caused due to deficiency of:

a. Iodine b. Calcium c. Iron d. Phosphorus

22. Which hormone helps in development of female secondary sexual characteristics?

a. Estrogen b. Testosterone c. Thyroxine d. Adrenaline

23. A neuron transmits messages in the form of:

a. Chemical energy b. Magnetic impulses c. Electrical impulses d. Sound waves

24. Which plant hormone is responsible for bending of shoots toward light?

a. Gibberellin b. Auxin c. Cytokinin b. Abscisic acid

25. Which is the fastest acting system for coordination?

a. Nervous system b. Muscular system c. Skeletal system d. Circulatory system

Section B: High Order Thinking Questions:

1: Imagine you accidentally touch a hot plate. You immediately pull your hand back. What is the correct sequence of events in this reflex action?

(A) Receptor in the skin →Spinal cord → Motor neuron → Sensory neuron →Effector muscle

(B) Receptor in the skin →Sensory neuron →Spinal cord → Motor neuron→ Effector muscle

(C) Spinal cord → Sensory neuron →Receptor in the skin →Motor neuron → Effector muscle

(D) Effector muscle → Motor neuron → Spinal cord → Sensory neuron → Receptor in the skin

2: A gardener notices that his sunflower plants are all facing the sun throughout the day. Which plant hormone is primarily responsible for this directional growth response?

(A) Gibberellins (B) Cytokinins (C) Abscisic acid (D) Auxins

3: You are running a race. Your heart rate increases, you breathe faster, and more blood is directed to your muscles. Which part of your nervous system is primarily responsible for these "fight or flight" responses?

(A) Somatic nervous system (B) Sympathetic nervous system (C) Parasympathetic nervous system (D) CNS

4: A doctor taps your knee with a hammer, and your lower leg jerks forward. This is a knee-jerk reflex. Which of the following is NOT involved in this reflex arc?

(A) Brain (B) Sensory neuron (C) Spinal cord (D) Motor neuron

5: A student observes a Mimosa (touch-me-not) plant. When she touches its leaves, they fold up and droop. This movement is a response to touch. What type of movement is this?

(A) Phototropism (B) Geotropismm (C) Thigmonasty (D) Chemotropism

6: A person is having difficulty maintaining their balance while walking. A doctor suspects that a specific part of their brain is not functioning properly. Which part of the brain is most likely affected?

(A) Cerebrum (B) Medulla (C) Cerebellum (D) Hypothalamus

7: A botanist is trying to promote the growth of lateral buds in a plant. Which plant hormone should she apply to the plant to achieve this?

(A) Auxins (B) Gibberellins (C) Cytokininsm (D) Abscisic acid

8: You are walking in a dark room and suddenly your pupils dilate to let in more light. Which part of your brain controls this involuntary action?

(A) Forebrain (B) Midbrain (C) Hindbrain (D) Cerebrum

9: A person has a problem with their thyroid gland and is experiencing a slow metabolism and weight gain. Which hormone is likely being secreted in insufficient amounts?

(A) Insulin (B) Adrenaline (C) Thyroxine (D) Growth hormone

10: A scientist is studying a plant and observes that its roots are growing downwards into the soil. This response is known as:

(A) Phototropism (B) Hydrotropism (C) Geotropism (D) Chemotropism

11: During a medical examination, a doctor shines a light into a patient's eye, and the pupil constricts. Which part of the nervous system is responsible for this reflex action?

(A) Somatic nervous system (B) Autonomic nervous system (C) Central nervous system (D) Peripheral nervous system

12: A person is suffering from diabetes mellitus. This condition is caused by a deficiency of which hormone produced by the pancreas?

(A) Thyroxine (B) Adrenaline (C) Growth hormone (D) Insulin

13: A farmer sprays his crops with a hormone that promotes cell division and breaks the dormancy of seeds. Which plant hormone is he most likely using?

(A) Auxins (B) Gibberellins (C) Cytokinins (D) Abscisic acid

14: You are eating a delicious meal. The salivation in your mouth, the release of digestive juices, and the peristalsis in your intestines are all controlled by which part of your nervous system?

(A) Sympathetic nervous system (B) Somatic nervous system (C) Parasympathetic nervous system (D) CNS

15: A plant is kept in a dark room. The stem grows towards a small light source from a window. Which of the following statements about this response is true?

(A) The plant is showing negative phototropism. (B) Auxins are uniformly distributed throughout the stem.

(C) Auxins accumulate on the shaded side of the stem, causing faster cell elongation there.

(D) The plant's roots will also grow towards the light.

16: A person is unable to produce enough vasopressin (ADH). What is the likely consequence of this deficiency?

(A) Increased blood pressure (B) Reduced water absorption by the kidneys, leading to increased urination

(C) Increased glucose levels in the blood (D) Reduced metabolic rate

17: A student is performing an experiment where he places a plant on its side. After a few days, he observes that the stem bends upwards. This response is known as:

(A) Positive phototropism (B) Negative geotropism (C) Positive hydrotropism (D) Negative chemotropism

18: A young child is experiencing dwarfism. A doctor might suggest that there is a deficiency in the secretion of which hormone?

(A) Thyroxine (B) Adrenaline (C) Growth hormone (D) Insulin

19: You are trying to find something in a very cluttered drawer. The conscious effort of thinking, planning, and remembering where you last saw the item is primarily controlled by which part of your brain?

(A) Cerebellum (B) Medulla (C) Hypothalamus (D) Cerebrum

20: A scientist is trying to inhibit the growth of plants in a specific area. Which plant hormone should she use to cause wilting and dormancy?

(A) Auxins (B) Gibberellins (C) Cytokinins (D) Abscisic acid

Section c: Assertion Reason Type Questions:

Instructions: For each question, select the correct option:

(A) Both assertion and reason are true, and reason is the correct explanation of the assertion
(B) Both assertion and reason are true, but reason is not the correct explanation of the assertion
(C) Assertion is true, but reason is false **(D)** Assertion is false, but reason is true

Q1.**Assertion:** Reflex actions are controlled by the spinal cord.
Reason: The brain does not participate in any reflex action.

Q2. **Assertion:** The cerebellum helps in maintaining posture and balance.
Reason: The cerebellum controls voluntary functions of internal organs.

Q3. **Assertion:** Auxins promote bending of stems towards light.
Reason: Auxins are distributed equally on both sides of the stem.

Q4. **Assertion:** Adrenaline increases heart rate during stressful conditions.
Reason: Adrenaline prepares the body for fight or flight response.

Q5. **Assertion:** Thyroxine regulates carbohydrate, fat, and protein metabolism.
Reason: Thyroxine is secreted by the pituitary gland.